MIL-R-44023A(GL)
7 May 1981
SUPERSEDING
MIL-R-44023(GL)
26 March 1980

MILITARY SPECIFICATION

REPRODUCTION SET, DIAZO PROCESS, FOR FIELD USE

This specification is approved for use by the Natick Research and Development Laboratories, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of diazo process reproduction set for field use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document of by letter.

FSC 3610

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DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

5-7-173	-	Reproduction Set, Diazo Process, For Field Use; Carrying Case for Chart/Overlay Copier; Assembly
5-7-174	-	Cover
5-7-175	-	Base
5-7-176	-	Support Assembly
5-7-177	-	Cover Gasket
5-7-178	-	Base Gasket
5-7-179	-	Strap Assembly
5-7-180	-	Back Up Plates
5-7-181	-	Carrying Case for Chart/Overlay Copier Expendable Supplies Assembly
5-7-182	-	Case Assembly
5-7-183	-	Cover Assembly
5-7-184	-	Tray
5-7-185	-	Angles and Back Up Plates
5-7-186	-	Hinge
5-7-187	-	Container Box for Operating Supplies

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS LABORATORIES INC.

UL 114 - Standard for Office Appliances and Business Equipment

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

3. REQUIREMENTS

3.1 Alternate components. Components offered as equivalent to components specified herein and on the referenced drawings as a specific manufacturer's part number or equal shall be functionallly equal, of equal or better quality with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance

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to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

* 3.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

* 3.2.1 UL compliance and certification. The reproduction chart overlay copier machines shall comply with the applicable requirements of UL 114. Prior to the first shipment, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the machines furnished under this specification meet the applicable UL requirements. Acceptable evidence of meeting the requirements of UL shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory, approved by the contracting officer, indicating that the machines have been tested and conform to UL 114.

3.3 Materials and components. Materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed items comply with all the provisions of this specification (see 6.3).

3.4 Design and construction. The design and construction of the reproduction sets shall be as specified herein and as shown on the drawings listed in 2.1.

* 3.4.1 Copier, chart overlay and carrying case. The chart and overlay copier shall be a table top mounted machine with a printing size capacity of 48 inches. The machine shall consist of a feedboard, monofilament paper cutter, four fluorescent lamps, solid state electronic controls, reverse switch, hi-lo switch and a dual power connection to permit operation at 120 volts, 60 hertz, single phase or 220 volts, 50 hertz, single phase. The developer section shall use ammonia vapor with heaters in a closed system and shall be ventless. The copier shall operate at speeds up to 25 feet/minute continuously, without damage or malfunction, when tested as specified in 4.5.1. The copier shall be capable of reproducing original transparent overlays of a grease pencil acetate, when tested as specified in 4.5.4. The copier shall fit inside the carrying case. The case shall be 17 inch high x 70 inch long x 26 inch deep. The case with copier shall be capable of being hand carried by four persons without failure of handles, straps or latches when tested as specified in 4.5.2.

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* 3.4.2 Supplies, expendable and carrying case. The expendable supplies shall be furnished in accordance with table I. The expendable supplies shall fit inside the expendable supply case. The expendable supply case shall be 10 inch high x 37 inch long x 24 inch deep. The case with supplies shall be capable of being hand carried by two persons without failure of handles or latches when tested as specified in 4.5.3.

TABLE I. Supplies, expendable

Item	Description	Quantity
1	Paper, copying, diazo process, standard weight (20 lb), sensitized one side, rapid speed, blackline, 36 inch wide x 50 yards Part No. 108S, GAF Corporation or equivalent	1 roll
2	Paper, copying, diazo process, standard weight (20 lb), sensitized one side, rapid speed, blueline, 22 inch x 34 inch 250 sheets/package, Part No. 208S, GAF Corporation or equivalent	1 package
3	Paper, Copying, diazo process, sepia image, transparentized, high strength base paper, rapid speed, 36 inch wide x 50 yards Part No. 405ZT, GAF Corporation or equivalent	1 roll
4	Film, diazo type (clear base), standard speed, density (0.005 inch thick), black image, 36 inch wide x 20 yards Part No. 102V, GAF Corporation or equivalent	1 roll
5	Film, diazo type (overlay sheets), standard speed, density (0.005 inch thick), black image, 22 inch x 34 inch 25 sheets/package, Part No. 102V, GAF Corporation or equivalent	4 packages
6	Film, diazo type (overlay sheets) standard speed, density (0.005 inch thick), black image, 8-1/2 inch x 11 inch 100 sheets/ package, Part No. 102V, GAF Corporation or equivalent	3 packages
7	Film, diazo type intermediate intensifying, polyester, 22 inch x 34 inch (0.0029 inch total thickness) 100 sheets/package, Part No. 501P2, GAF Corporation or equivalent	2 packages

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* 3.4.3 Supplies, operating and boxes. The operating supplies shall be furnished in accordance with table II. The supplies shall fit in three boxes.

TABLE II. Supplies, operating

Item	Description	Quantity
1	Absorber, container, 9 inch wide x 9-1/4 inch deep x 10 inch high or charcoal filter, as applicable	3 (1 each per box)
2	Ammonium hydroxide, 1 gallon 26° Baume, Part No. 892-045 GAF Corporation or equivalent	6 (2 gallons each per box)

* 3.4.4 Spare parts. The spare parts shall be furnished in accordance with table III, and they shall fit inside the copier carrying case.

TABLE III. Spare parts

Item	Description	Quantity
1	Fuse, machine main	2 each
2	Fuse, machine drive	2 each
3	Lamp, Fluorescent	1 each

3.5 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.6 Marking. The copier machine, copier carrying case and expendable supply carrying case shall be marked for identification in accordance with the manufacturer's commercial practice and shall include the item identification, stock No., manufacturer's name and date of manufacture.

3.7 Workmanship. The units shall be free from defects, such as fractures, splits, punctures, tears, dents, creases, deterioration or malformation. There shall be no sharp edges, slivers, burrs or projections. The finish shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch-up, scale, scratches, stain, overspray, sag, imprint or embedded foreign matter. Threaded fasteners shall not be broken, stripped, fractured or loose. Rivets shall completely fill the holes, with full size heads neatly finished and in full contact with the surface of the riveted members and shall not be loose. Wiring shall not be cut, abraded or have insulation stripped, and shall be joined at the terminals. Wiring shall have adequate slack to provide strain relief.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined as specified in 4.4.2.1 and 4.4.2.2 and tested as specified in 4.5.1 thru 4.5.4. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item inspection. The inspection lot shall be all reproduction sets offered for inspection at one time. The sample unit shall be one complete assembled reproduction set.

4.4.2.1 Visual examination. Examination shall be made of the end item listed in tabl IV. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE IV. Visual examination

Examine	Defect	Classification Major	Classification Minor
Finish	Color not as specified		X
	Blistered, peeled, chipped or area of no film		X
	Poor touch-up, scale, scratched, stained, overspray, sag, imprint or foreign matter embedded		X
Design, construction and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size	X	
	Fractured, split, punctured, torn, dented, creased, deteriorated or malformed	X	
	Component loose		X
	Sharp burr or edge or sliver	X	
	Any functioning component that is inoperative or will not operate as intended	X	
	Any adjustable assembly that cannot be adjusted		X
	Any assembly not properly adjusted		X
	Any component not readily accessible for servicing where required	X	
Bolts, nuts, screws, studs and other types of threaded fasteners	Broken, stripped, fractured or not tight		X
Rivets	Broken, not peened or insufficiently peened or loose		X
Electrical	Wiring cut, abraded, or not properly installed or joined	X	
	Adequate slack not provided for wiring to relieve strain	X	
Marking	Missing, incomplete, not legible		X

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4.4.2.2 Dimensional examination. Examination shall be made of the reproduction set for compliance with dimensions specified. The inspection level shall be S-2, with an AQL of 4.0, expressed in terms of defects per hundred units. Any nonconformance shall be classified as a defect.

* 4.4.2.3 Testing of the end item. Each end item shall be tested as specified in 4.5.1 and 4.5.4. Failure to pass either test shall be cause to reject the effected end item.

4.4.3 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with section 5. Defects shall be as listed in table V. The sample unit shall consist of one reproduction set fully packaged. The lot shall be the number of units offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

TABLE V. Packaging inspection

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing or damaged
Workmanship	Inadequate cushioning

* 4.4.4 Certificate examination. Certificates of compliance, certified test reports, or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated (see 3.2.1).

4.5 Methods of inspection.

* 4.5.1 Copier, speed test. The copier machine shall be run continuously with speeds up to 25 feet/minute, when in the 120 volts, 60 hertz mode, to reproduce sixty prints of size E military drawing without stopping. Any evidence of damage or malfunction shall constitute failure of this test (see 3.4.1).

4.5.2 Copier, carrying case, test. The copier mounted in the carrying case shall be lifted by the four side handles and transported by four persons for a minimum distance of 50 feet. Failure of any handle, strap, or latch to support the load shall constitute failure of this test (see 3.4.1).

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4.5.3 Expendable supplies, carrying case, test. The expendable supplies case loaded with expendable supplies shall be lifted by two persons and transported a minimum distance of 50 feet. Failure of any handle or latch to support the case shall constitute failure of this test (see 3.4.2).

* 4.5.4 Reproduction test. The copier shall be run at a slow speed to reproduce 15 (36 x 48 inch) transparent overlays from an intermediate film original. The intermediate film original shall be made from a grease pencil clear acetate original. Any evidence of damage or malfunction shall constitute failure of this test (see 3.4.1).

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. All the components of each reproduction set shall be placed into proper carrying cases or boxes. Cushioning shall be used, as necessary, to prevent movement or damage during transit. Closure of the carrying case and boxes shall be accomplished by means provided.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each complete reproduction kit, preserved as specified in 5.1, shall be packed in a cleated-plywood shipping container conforming to overseas type, style A of PPP-B-601. Cushioning shall be used, as necessary, to fill voids within the shipping container and prevent movement or damage of contents during transit. Each shipping container shall be closed and strapped in accordance with PPP-B-601.

5.2.2 Commercial packing. Reproduction kits, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The items covered are intended to be used for the reproduction of overlays, operation orders, tactical map overprints, situation maps and sketches by Army units in tactical operations.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.2, 4.3 and 6.4).
(c) Contract data requirements (see 3.5).
(d) Selection of applicable level of packing (see 5.2).

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6.2.1 Contract data requirements. Any data item to be delivered under any contract for items should be specifically called for in the contract in accordance with the applicable regulation of the procuring activity (see 3.5 and 6.2).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

* 6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the DAR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-44023A(GL) Reproduction Set, Diazo Process, For Field Use

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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